

```
In [1]: import zipfile

with zipfile.ZipFile("ml_app..zip", "r") as zip_ref:
    zip_ref.extractall("ml_app_data")
```

```
In [2]: import os

os.listdir("ml_app_data")
```

```
Out[2]: ['insurance.csv']
```

```
In [3]: import pandas as pd

df = pd.read_csv("ml_app_data/insurance.csv")

df.head()
```

```
Out[3]:
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
In [4]: from sklearn.ensemble import RandomForestRegressor
```

```
In [6]: from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()

df["sex"] = le.fit_transform(df["sex"])
df["smoker"] = le.fit_transform(df["smoker"])
df["region"] = le.fit_transform(df["region"])

df.head()
```

```
Out[6]:
```

	age	sex	bmi	children	smoker	region	charges
0	19	0	27.900	0	1	3	16884.92400
1	18	1	33.770	1	0	2	1725.55230
2	28	1	33.000	3	0	2	4449.46200
3	33	1	22.705	0	0	1	21984.47061
4	32	1	28.880	0	0	1	3866.85520

```
In [8]: X = df.drop("charges", axis=1)
        y = df["charges"]
```

```
In [18]: from sklearn.model_selection import train_test_split

        X_train, X_test, y_train, y_test = train_test_split(
            X,
            y,
            test_size=0.2,
            random_state=42
        )
```

```
In [12]: from sklearn.ensemble import RandomForestRegressor

        model = RandomForestRegressor(
            n_estimators=100,
            random_state=42
        )

        model.fit(X_train, y_train)
```

```
Out[12]: ▼ RandomForestRegressor ⓘ ?
        ► Parameters
```

```
In [15]: from sklearn.metrics import r2_score

        pred = model.predict(X_test)

        print("R2 Score:", r2_score(y_test, pred))
```

R2 Score: 0.8642606273649586

```
In [11]: %%writefile untitled.py

import streamlit as st
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestRegressor

# Load dataset
df = pd.read_csv("ml_app_data/insurance.csv")

df["sex"] = LabelEncoder().fit_transform(df["sex"])
df["smoker"] = LabelEncoder().fit_transform(df["smoker"])
df["region"] = LabelEncoder().fit_transform(df["region"])

X = df.drop("charges", axis=1)
y = df["charges"]

model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X, y)

st.title("Medical Insurance Cost Predictor")
```

```
age = st.number_input("Age", 1, 100, 25)

sex = st.selectbox(
    "Sex",
    ["female", "male"]
)

bmi = st.number_input(
    "BMI",
    min_value=10.0,
    max_value=60.0,
    value=25.0
)

children = st.number_input(
    "Children",
    min_value=0,
    max_value=10,
    value=0
)

smoker = st.selectbox(
    "Smoker",
    ["no", "yes"]
)

region = st.selectbox(
    "Region",
    ["northeast", "northwest", "southeast", "southwest"]
)

sex = 1 if sex == "male" else 0
smoker = 1 if smoker == "yes" else 0

region_map = {
    "northeast": 0,
    "northwest": 1,
    "southeast": 2,
    "southwest": 3
}

region = region_map[region]

if st.button("Predict"):

    input_data = pd.DataFrame(
        [[age, sex, bmi, children, smoker, region]],
        columns=[
            "age",
            "sex",
            "bmi",
            "children",
            "smoker",
            "region"
        ]
    )
```

```
)  
  
prediction = model.predict(input_data)  
  
st.success(  
    f"Estimated Insurance Cost: ${prediction[0]:,.2f}"  
)
```

Overwriting untitled.py

```
In [7]: %%writefile untitled.py  
  
import streamlit as st  
  
st.title("Medical Insurance Cost Predictor")  
st.write("Test App Running")
```

Overwriting untitled.py

```
In [8]: with open("untitled.py", "r") as f:  
        print(f.read())
```

```
import streamlit as st
```

```
st.title("Medical Insurance Cost Predictor")  
st.write("Test App Running")
```

```
In [ ]:
```

```
In [ ]:
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